Machine Learning for Fault Detection and Diagnosis in the Tennessee Eastman Process (TEP)

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Abstract

Reliable fault detection and diagnosis is important for complex chemical processes, where a fault might be captured from evolving patterns across multiple variables. This study investigates multi-class fault diagnosis using machine learning techniques on the classic Tennessee Eastman Process (TEP) simulation dataset. Out of the 21 fault scenarios, 4 classes are selected: Normal operation, Fault 1, Fault 2, and Fault 4. Logistic Regression and Random Forest are first established as baselines, their results motivating further investigation of some informative process variables and patterns. After some investigation, a one-dimensional convolutional neural network (1D CNN) and a gated recurrent unit (GRU) network are developed, learning from temporal patterns of data. Model performances are evaluated using prediction accuracy, macro-averaged F1-score, confusion matrices, and fault-specific detection delay.

From the experimental results, Logistic Regression achieves a sample-level test accuracy of 99.09%, while Random Forest achieves 97.89%. The CNN reaches 99.35% sample-level accuracy after window predictions are aggregated, compared with 99.22% for the GRU. However, the models exhibit different detection delays across fault classes, indicating that overall classification metrics alone do not fully characterize temporal detection behavior. These findings demonstrate the value of evaluating both classification performance and fault-onset response.

1. Introduction

1.1 Background and Objectives

Modern chemical processes involve multivariable control systems and dynamic relationships among process variables. Changes in operating conditions can propagate through a process, and even a small fault in one may lead to overall severe degradation (Muhammad et al., 2021). Hence reliable and timely identification of faults is essential. One appealing approach to this problem is machine learning.

The Tennessee Eastman Process (TEP), a plant-wide process control problem by Downs and Vogel (1993), is a widely recognized benchmark for machine learning experiments with Fault Detection and Diagnosis (FDD) methods (Golyadkin et al., n.d.). Its simulated chemical process provides a multivariable environment in which different operating conditions can be studied systematically.

This study investigates the effectiveness of machine learning-based multi-fault diagnosis on 4 selected fault cases from the classic TEP dataset: Normal Operation, Fault 1, Fault 2, and Fault 4.

The investigation begins with Logistic Regression and Random Forest as classical machine learning baselines. Random Forest feature importance is then examined to identify process variables that may be informative for distinguishing operating conditions.

Temporal patterns of the variables are observed, motivating the subsequent investigation of models that learn from temporal patterns in data: a one-dimensional convolutional neural network (1D CNN) and a gated recurrent unit (GRU) network.

Their performance is compared with that of the classical baselines, with particular attention to consistent sample-level evaluation and fault detection delay.

2. Dataset and Problem Formulation

2.1 Tennessee Eastman Process Dataset

This study uses a TEP dataset provided by The Board of Trustees of the University of Illinois. Each observation, representing a simulated process measurement taken every 3 minutes, contains 52 process variables:

- 41 measured process variables, denoted by xmeas_1 to xmeas_41.
- 11 manipulated variables, denoted by xmv_1 to xmv_11.

This study is restricted to 4 cases: Normal Operation, Fault 1, Fault 2, and Fault 4. It does not attempt to diagnose all of the 21 faults represented in the complete TEP benchmark. The classes are encoded using labels (0), (1), (2), and (4), respectively. The numerical labels identify classes and do not imply an ordinal relationship.

2.2 Training, Validation, and Test Data

The training dataset is constructed from the training sequences for Normal Operation and Faults 1, 2, and 4. The resulting dataset contains 1,940 observations and 52 features per observation.

The training data are divided chronologically into fitting and validation subsets, with an 80:20 split applied separately to each sequence. The earlier portion is used for model fitting, while the later portion is reserved for validation. The chronological splitting is to preserve the time order; if random splitting is used, the model may train on later observations and validate on earlier observations for example.

The resulting fitting subset contains 1,552 observations, and the validation subset contains 388 observations.

The test dataset combines four test sequences, each containing 960 observations. For the three fault test sequences, the TEP convention is that the first 160 observations represent normal operation before the fault onset (Zhang et al., 2021), followed by the faulty operating period. The combined test dataset therefore contains 3,840 observations:

Class	Test observations
Normal	1,440
Fault 1	800
Fault 2	800
Fault 4	800
Total	**3,840**

The test sequences preserve the transition from normal operation to faulty operation, allowing evaluation of both classification performance and prediction behavior around fault onset.

3. Methodology

3.1 Classical Machine Learning Baselines

Two classical models are used to establish a reference against which more flexible nonlinear and temporal models can be compared.

3.1.1 Logistic Regression

Logistic Regression estimates class probabilities from a weighted combination of input features and assigns each observation to the predicted class.

The implementation uses feature standardisation (because input variables may have different numerical scales) followed by a Logistic Regression classifier with a maximum of 2,000 iterations and a fixed random seed.

3.1.2 Random Forest

Random Forest is an ensemble of decision trees that can learn nonlinear relationships and interactions among process variables.

The implementation uses 200 trees, a fixed random seed, and a minimum of two samples per leaf. The model is trained on the same multi-class fitting data used for the baseline comparison.

3.2 Exploratory Analysis of Process Variables

Following baseline evaluation, Random Forest feature importance is used to identify to what extent each variable contributes to the model's predictions. Potentially informative process variables are examined across the four operating conditions using numerical summaries and time-series visualizations.

The time-series plots also motivate investigation of temporal models, as some fault-related patterns appear to evolve rather than emerge as an instantaneous, constant shift.

3.3 One-Dimensional Convolutional Neural Network

A one-dimensional CNN (1D CNN) is developed to classify temporal windows of multivariate process data. It extracts relevant features from sequential data using convolutional filters (windows) that slide along a single dimension. The extracted features are then used for classification or regression tasks, making 1D CNNs suitable for time-series analysis, signal processing and so on.

The model is trained using labelled windows constructed from the fitting sequences. Each input window contains 50 time steps and 52 variables. There is a stride of 5 samples between successive windows. Validation data are used to monitor training behavior and assess performance during development.

3.4 Gated Recurrent Unit Network

A Gated Recurrent Unit (GRU) network is introduced as an alternative temporal architecture. A GRU is a type of recurrent neural network used to analyze sequential data. It uses two gates, called the update gate and reset gate, to control how information from previous time steps is retained or forgotten. This enables the GRU to learn relationships between data points across different time steps, making it suitable for modeling temporal dependencies in sequential data.

The implemented GRU is trained using sliding windows of 50 consecutive observations and 52 variables, with a stride of 5. The model uses an input dimension of 52, a hidden dimension of 32, one recurrent layer, and a four-class output layer. The model contains approximately 8,388 trainable parameters.

Training uses the Adam optimizer with a learning rate of 0.001, cross-entropy loss, a batch size of 32, and a maximum of 50 epochs. The same 50-step window representation is used for the temporal comparison.

3.5 Evaluation and Detection Delay

Model performance is evaluated using accuracy, macro-averaged F1-score, class-wise metrics, and confusion matrices.

For 1D CNN and GRU, because consecutive windows overlap, a single sample receives predictions from multiple windows. Hence, the evaluation proceeds at two levels for the temporal models:

- First, window level metrics quantify how accurately each model classifies temporal windows.
- Second, predictions from overlapping windows are combined for each sample to obtain sample-level predictions. This allows direct comparison with the classical models.

Detection delay is then computed for each fault. For a fault with onset index (t_0), the delay is the number of samples between onset and the first point at which three consecutive predictions correctly identify the fault.

This metric complements accuracy and F1-score. A model may achieve high overall accuracy but still take longer to identify a particular fault after it begins.

The test results are interpreted in the context of the selected sequences and experimental protocol.

4. Results

4.1 Classical Baseline Performance

The classical models achieve high classification performance on the combined test dataset.

Model	Accuracy	Macro F1	Misclassified samples
Logistic Regression	99.09%	99.15%	35
Random Forest	97.89%	98.06%	81

Logistic Regression outperforms Random Forest on both reported metrics. This result suggests that the selected operating conditions are highly distinguishable using the available features and that a linear model provides a strong baseline. Random Forest nevertheless provides useful nonlinear modeling capacity and feature-importance estimates for the subsequent exploratory analysis.

```
Confusion matrix:
[[1373    0   67    0]
 [   2  798    0    0]
 [  12    0  788    0]
 [   0    0    0  800]]
```

	Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
0	Logistic Regression	0.9909	0.9918	0.9911	0.9915
1	Random Forest	0.9789	0.9779	0.9840	0.9806

The confusion matrices indicate that Fault 4 is classified correctly in the recorded results for both classical models. A notable difference concerns normal observations: Random Forest misclassifies more normal samples as Fault 2 than Logistic Regression does. This contributes to its lower overall performance.

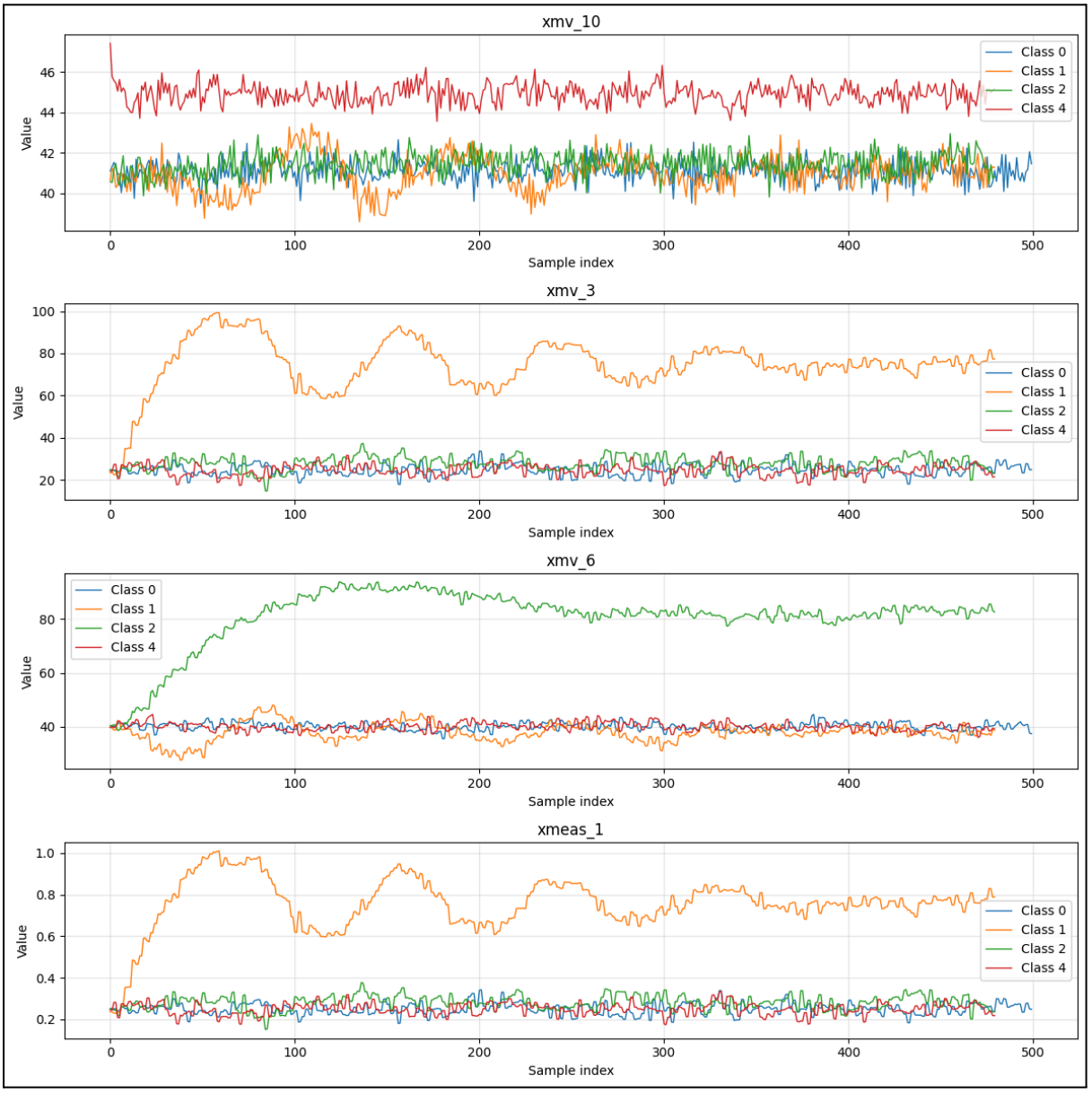
The results demonstrate that a more complex model does not necessarily produce better classification performance on every dataset. They also motivate investigation of which process variables distinguish the selected conditions.

4.2 Predictive Relevance of Process Variables

Random Forest feature importance identifies xmv_10 as the most important variable in the recorded analysis, followed by xmeas_1, xmv_6, xmv_3, and xmeas_18.

Variable	Recorded importance
xmv_10	0.2557
xmeas_1	0.0716
xmv_6	0.0699
xmv_3	0.0649
xmeas_18	0.0582

The relatively high importance of xmv_10 suggests that it is particularly informative for distinguishing the selected operating conditions in the fitted Random Forest model. The prominence of xmeas_1, xmv_6, xmv_3, and other measured variables suggests that multiple process variables contribute to classification.



Subsequent numerical and time-series visualisation analyses identify candidate signatures associated with the selected fault classes. xmv_3 and xmeas_1 are associated with Fault 1, while xmv_6 exhibit informative patterns for Fault 2. xmv_10 is particularly informative for Fault 4.

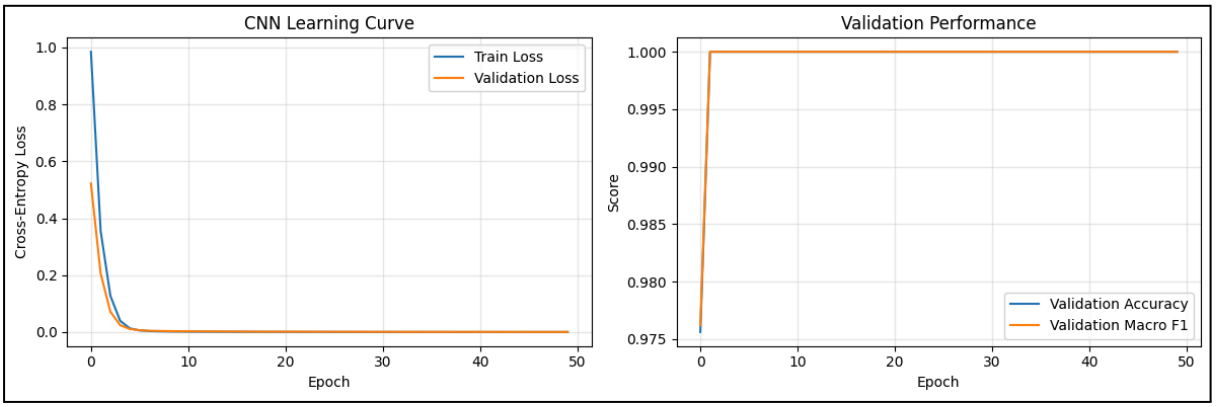
These observations are useful for understanding the model's predictions and motivating temporal modeling.

4.3 CNN Window-Level Performance

The CNN is evaluated using sliding windows of 50 observations with a stride of 5 samples. The recorded test set contains 732 windows.

Model	Window-level accuracy	Window-level macro F1
CNN	99.09%	99.15%

The CNN achieves high window-level performance, indicating that the learned representations distinguish the four classes effectively under the selected evaluation conditions.



The learning curves indicate that the model reaches strong performance relatively quickly. This may be consistent with the presence of class-specific patterns in the input data, although the current experiments do not isolate the contribution of temporal information from other factors.

Window-level metrics alone are not directly comparable with the sample-level metrics of the classical models. Therefore, window predictions are aggregated before drawing conclusions about relative model performance.

4.4 Sample-Level Comparison

After aggregation of the CNN's window predictions, its performance can be compared with the classical models on the same 3,840 test observations.

Model	Sample-level accuracy	Macro F1	Errors
Logistic Regression	99.09%	99.15%	35
Random Forest	97.89%	98.06%	81
CNN	99.35%	99.39%	25

The CNN achieves the highest recorded sample-level accuracy and macro F1 among these three models. It makes 25 errors, compared with 35 for Logistic Regression and 81 for Random Forest.

```
Confusion matrix:
[[1440    0    0    0]
 [   5  795    0    0]
 [  20    0  780    0]
 [   0    0    0  800]]
```

The CNN confusion matrix shows that all normal test observations are classified correctly in the recorded output. However, five Fault 1 observations and 20 Fault 2 observations are classified as Normal. All 800 Fault 4 observations are classified correctly.

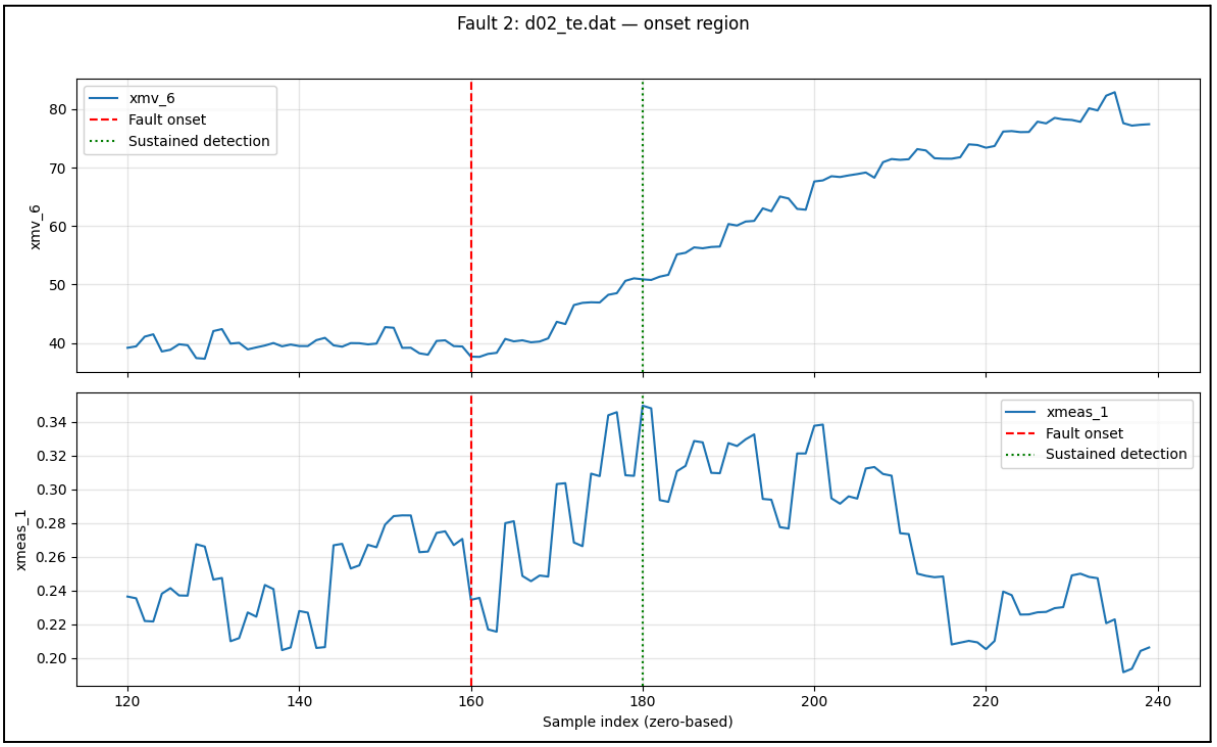
The remaining errors are therefore concentrated in Faults 1 and 2. This class-specific behavior motivates a closer examination of the temporal predictions around fault onset.

4.5 Fault Detection Delay

Detection delay is evaluated using the criterion of three consecutive correct predictions of the corresponding fault class.

Fault	CNN detection delay
Fault 1	5 samples
Fault 2	20 samples
Fault 4	0 samples

The recorded CNN results show different detection behavior across the three faults. Fault 4 is identified immediately under the selected criterion, while Fault 1 has a short delay. Fault 2 has the longest delay.



The Fault 2 result is consistent with the observation on some relevant variable patterns, particularly xmv_6 and xmeas_10, where the former evolves gradually after fault onset and the latter fluctuates all over the place without any specific pattern. The model may require additional observations before the pattern becomes sufficiently distinguishable for sustained correct classification.

4.6 CNN and GRU Comparison

The GRU is evaluated using the same temporal-window design. Its recorded window-level results are:

Model	Window-level accuracy	Window-level macro F1
CNN	99.45%	99.45%
GRU	99.32%	99.36%

The recorded sample-level results are:

Model	Sample-level accuracy	Macro F1	Errors
CNN	99.35%	99.39%	25
GRU	99.22%	99.27%	30

In these results, the CNN slightly outperforms the GRU on both sample-level metrics. However, detection delay varies by fault.

Fault	CNN delay	GRU delay
Fault 1	5 samples	0 samples
Fault 2	20 samples	10 samples
Fault 4	0 samples	20 samples

The GRU has shorter recorded delays for Faults 1 and 2, but a longer delay for Fault 4.

The results therefore do not support a universal conclusion that one temporal architecture is superior in every respect. The CNN achieves slightly higher overall classification metrics in the recorded experiment, while the GRU exhibits different fault-specific detection behavior.

5. Discussion

5.1 Classical Models as Strong Baselines

The high performance of Logistic Regression indicates that the selected operating conditions are distinguishable using the available process variables. This provides a useful baseline for evaluating the added complexity of temporal deep learning.

Random Forest performs less well in the recorded test results, despite its ability to represent nonlinear relationships. This highlights an important methodological point: model complexity alone does not guarantee better generalization.

5.2 From Feature Analysis to Temporal Modeling

Feature importance and time-series visualizations provide a motivation for temporal modeling. Some variables exhibit patterns that differ among operating conditions, and the changes may evolve across consecutive observations.

The CNN provides a way to learn local temporal patterns directly from multivariate windows, without requiring manual specification of all temporal features. Its high recorded classification performance is consistent with the possibility that these windows contain useful information.

5.3 Classification Performance and Detection Behavior

A central finding of the current analysis is that overall classification metrics and fault-specific detection delay provide complementary information.

The CNN achieves the strongest recorded sample-level accuracy among the compared models, but its delay is longer for Fault 2 than for the other two faults. The GRU has slightly lower aggregate classification metrics yet shorter recorded delays for Faults 1 and 2.

This distinction matters because a model's average performance does not fully describe its behavior during a transition from normal to faulty operation. The detection-delay metric helps characterize the onset response of the model, although it is not equivalent to a complete industrial alarm-performance evaluation.

The reported delays also depend on the chosen rule of three consecutive correct predictions. Different persistence criteria could produce different values, so the rule must remain explicit when comparing models.

6. Conclusion and Future Work

This study investigates multi-fault diagnosis using selected Tennessee Eastman Process data, comparing Logistic Regression and Random Forest baselines with a 1D CNN and a GRU network.

The recorded results show that classical models provide strong baselines, with Logistic Regression achieving 99.09% sample-level accuracy. The CNN achieves 99.35% sample-level accuracy and a macro F1-score of 99.39% after window predictions are aggregated. The GRU achieves 99.22% sample-level accuracy and a macro F1-score of 99.27% in the recorded experiment.

The comparison also demonstrates that aggregate classification metrics do not fully describe temporal detection behavior. The CNN and GRU exhibit different fault-specific detection delays, indicating that classification performance and onset response should be considered together when evaluating temporal fault diagnosis models.

The findings should be interpreted within the limits of the selected fault classes, simulated data, and existing experimental protocol.

Future work could extend the analysis in several directions. First, evaluation could include additional TEP fault classes and sequences reserved exclusively for final testing. Second, controlled experiments could investigate the contribution of temporal information by comparing window-based models with suitable non-temporal alternatives under identical data splits. Third, further work could examine the sensitivity of detection delay to the persistence criterion and evaluate false alarms, missed detections, and robustness to process variability. Finally, candidate fault signatures could be investigated with additional process knowledge to determine whether the observed predictive relationships have meaningful physical interpretations.

Overall, the study provides a structured comparison of classical and temporal machine learning approaches for selected TEP fault conditions. Its main methodological lesson is that fault diagnosis should be evaluated not only by how accurately a model classifies observations, but also by how its predictions behave as the process transitions from normal to faulty operation.

References

Database Used:
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